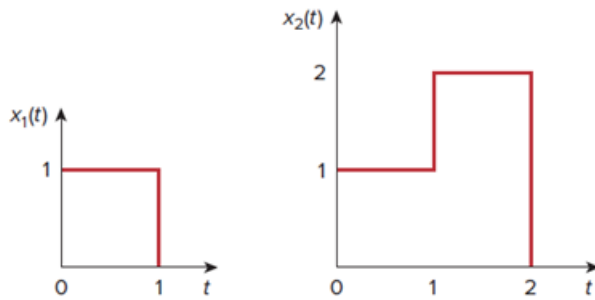


Problem

Graphically convolve the two functions in Fig. 15.14. To show how powerful working in the s -domain is, verify your answer by performing the equivalent operation in the s -domain.

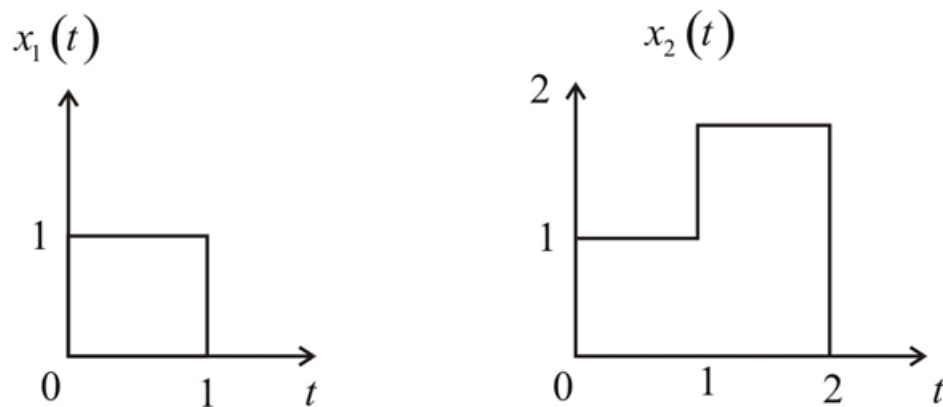
Figure 15.14:



Step-by-step solution

Step 1 of 12

Given two signals are

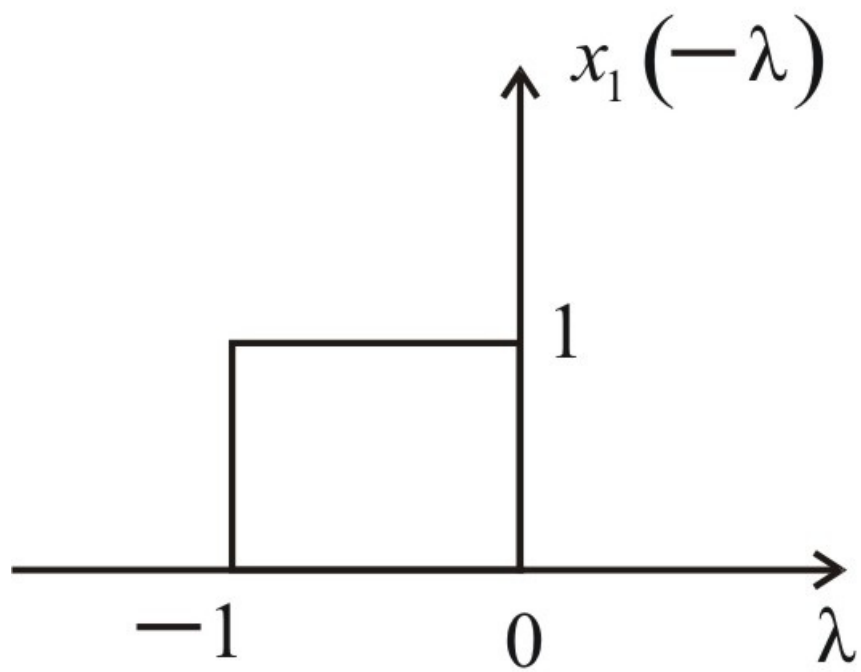


Step 2 of 12

To convolve the given two signals we follow the four steps to get $y(t) = x_1(t) * x_2(t)$

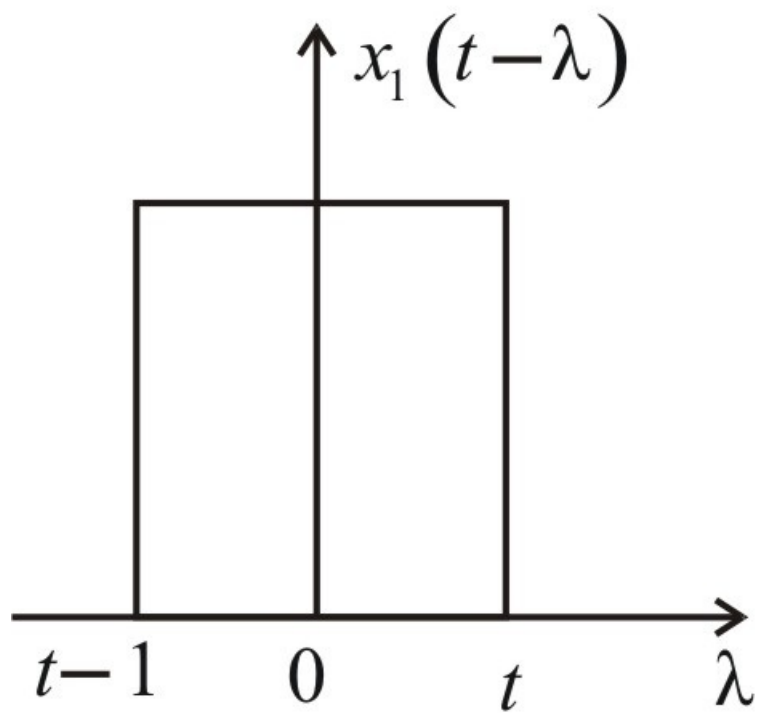
First, we fold $x_1(t)$ as

Step 3 of 12



Step 4 of 12

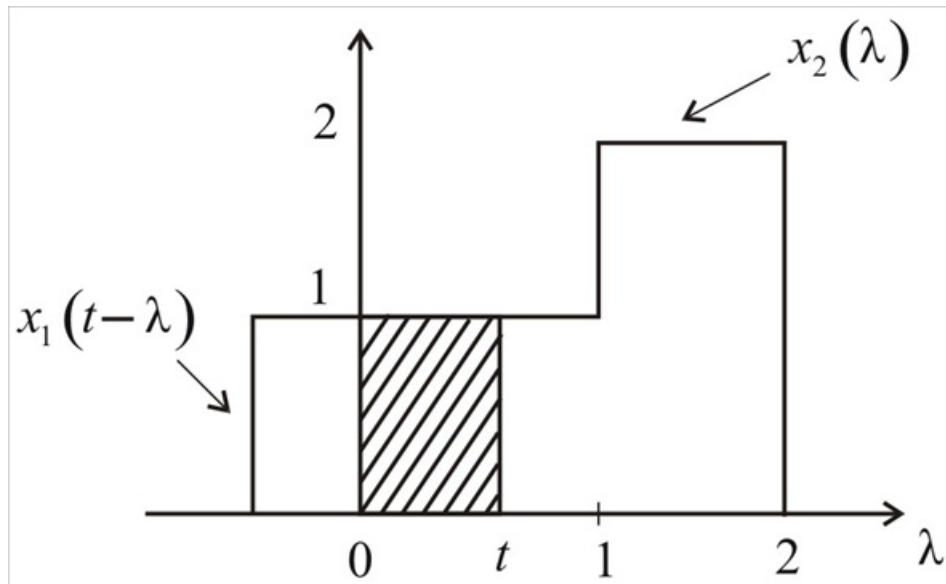
Now shift it by t as



Step 5 of 12

For different values of t , multiply the two functions and integrate to determine the area of the overlapping region.

For $0 < t < 1$, the two signals overlap between 0 and t as



Step 6 of 12

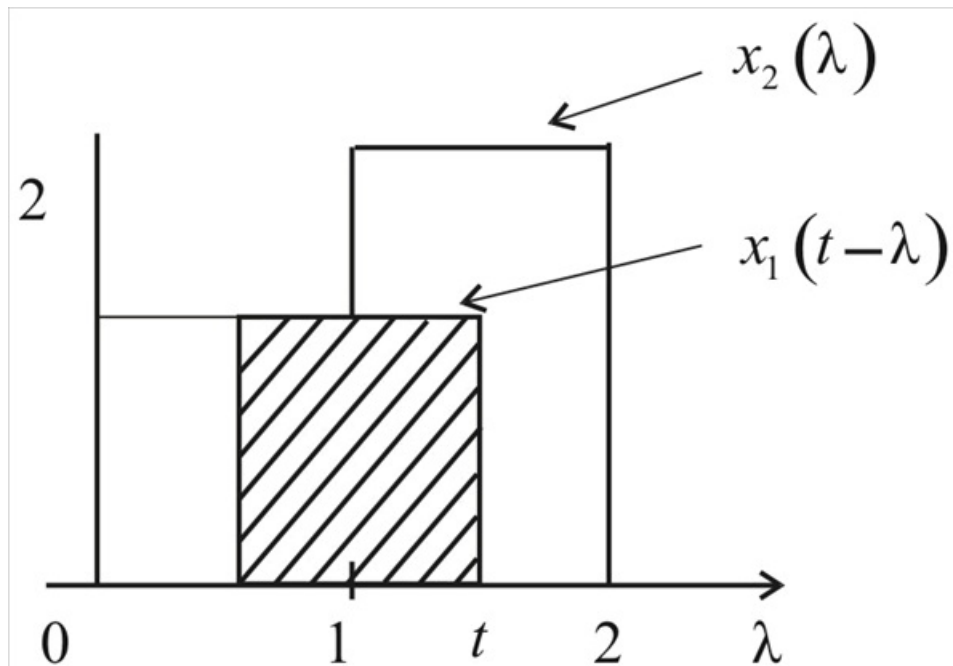
$$y(t) = \int_0^t (1)(1) d\lambda$$

$$y(t) = \lambda \Big|_0^t$$

$$y(t) = t \quad 0 < t < 1$$

Step 7 of 12

For $1 < t < 2$, the two signals overlap between $(t-1)$ to 1 and 1 to t as



Step 8 of 12

$$y(t) = \int_{t-1}^1 (1)(1) d\lambda + \int_1^t (2)(1) d\lambda$$

$$y(t) = \lambda \Big|_{t-1}^1 + 2\lambda \Big|_1^t$$

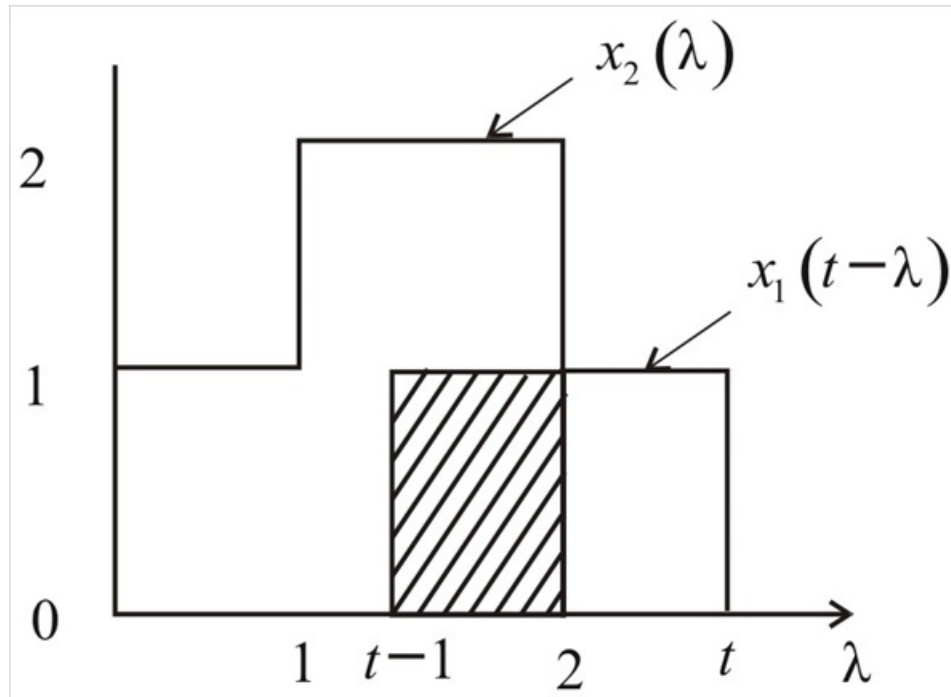
$$y(t) = 1 - t + 1 + 2(t - 1)$$

$$y(t) = 2 - t + 2t - 2$$

$$\therefore y(t) = t \quad 1 < t < 2$$

Step 9 of 12

For $2 < t < 3$, the two signals overlap between $(t-1)$ and 2,



$$y(t) = \int_{t-1}^2 (2)(1) d\lambda$$

$$y(t) = 2\lambda \Big|_{t-1}^2$$

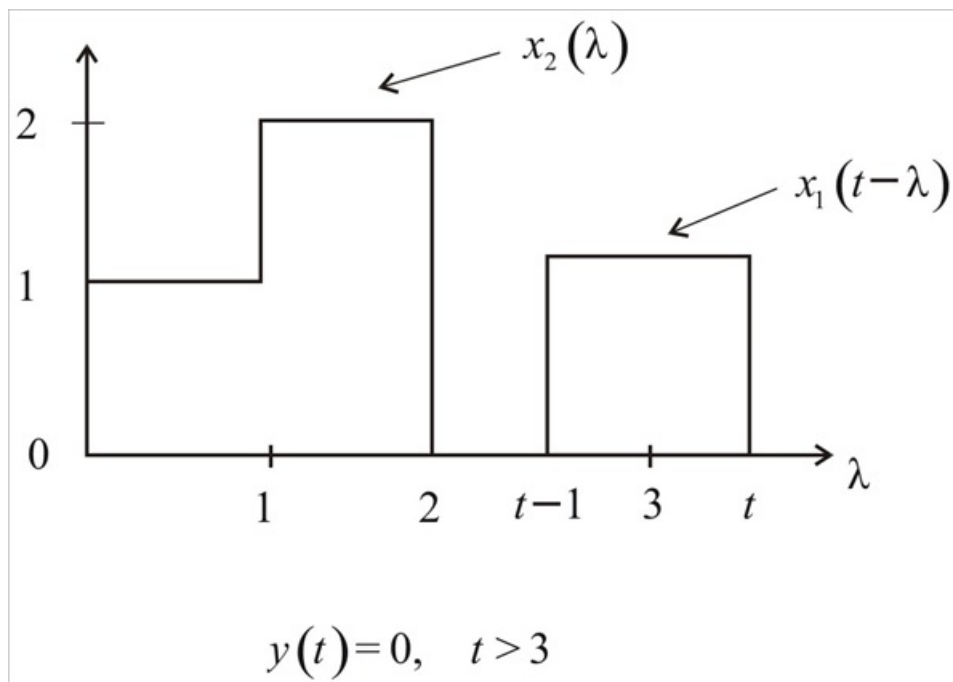
$$y(t) = 2(2 - t + 1)$$

$$y(t) = 2(2 - t)$$

$$\therefore y(t) = 6 - 2t \quad 2 < t < 3$$

Step 10 of 12

For $t > 3$, the two signals do not overlap



Step 11 of 12

Therefore, the result $y(t)$ is

$$y(t) = \begin{cases} t & 0 \leq t \leq 2 \\ 6-2t & 2 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Step 12 of 12

